# HW 2

## 1

### e

Our original LP is as follows:

$$\max x_1 + x_2$$

$$\text{s.t. } 2x_1 + x_2 \leq 3$$

$$0 \leq x_1, x_2 \leq 1$$

This is equivalent to:

$$\max x_1 + x_2$$

$$\text{s.t. } 2x_1 + x_2 \leq 3$$

$$x_1 \leq 1$$

$$x_2 \leq 1$$

$$0 \leq x_1, x_2$$

Thus the dual is:

$$\min 3y_1 + y_2 + y_3$$

$$\text{s.t. } 2y_1 + y_2 \geq 1$$

$$y_1 + y_3 \geq 1$$

$$y_1, y_2, y_3 \geq 0$$

Thus the complementary slackness conditions are as follows:

$$(2x_1 + x_2 - 3)y_1 = 0$$

$$(x_1 - 1)y_2 = 0$$

$$(x_2 - 1)y_3 = 0$$

$$(1 - 2y_1 - y_2)x_1 = 0$$

$$(1 - y_1 - y_3)x_2 = 0$$

### f

```python
from docplex.mp.model import Model

mdl = Model()

y1 = mdl.continuous_var(name='y1', lb=0)
y2 = mdl.continuous_var(name='y2', lb=0)
y3 = mdl.continuous_var(name='y3', lb=0)

mdl.add_constraint(2*y1 + y2 >= 1)
mdl.add_constraint(y1 + x3 >= 1)

mdl.minimize(3*y1 + y2 + y3)

solution = mdl.solve()

print(mdl.export_as_lp_string())

print(solution)
```

This results in a $y^*$ as follows:

$$y^* = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

We can substitute the values of $y^*$ to our slackness conditions:

$(2x_1 + x_2 - 3)0 = 0$

$\rightarrow 0 = 0$

$(x_1 - 1)1 = 0$

$\rightarrow x_1 = 1$

$(x_2 - 1)1 = 0$

$\rightarrow x_2 = 1$

$(1 - 0 - 1)x_1 = 0$

$$\rightarrow 0 = 0$$

$$(1 - 0 - 1)x_2 = 0$$

$$\rightarrow 0 = 0$$

Therefore we end up with $x^*, y^*$ as follows:

$$y^* = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, x^* = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

We can also see that by strong duality these solutions result in their objective values being 2 and equal. Thus these solutions are optimal.

# 3

## c

Our original LP is:

$$\min \gamma$$

$$\text{s.t. } 3x_1 + x_2 + x_3 + x_4 \leq \gamma$$

$$3x_1 + 4x_2 + 2x_3 + 2x_4 \leq 2\gamma$$

$$3x_1 + 4x_2 + 5x_3 + 3x_4 \leq 3\gamma$$

$$3x_1 + 4x_2 + 5x_3 + 6x_4 \leq 3\gamma$$

$$x_1 + x_2 + x_3 + x_4 = 1$$

$$x_1, x_2, x_3, x_4 \geq 0$$

Which can be rewritten as:

$$\max 0 + 0 + 0 + 0 - \gamma$$

$$\text{s.t. } 3x_1 + x_2 + x_3 + x_4 - \gamma \leq 0$$

$$3x_1 + 4x_2 + 2x_3 + 2x_4 - 2\gamma \leq 0$$

$$3x_1 + 4x_2 + 5x_3 + 3x_4 - 3\gamma \leq 0$$

$$3x_1 + 4x_2 + 5x_3 + 6x_4 - 3\gamma \leq 0$$

$$x_1 + x_2 + x_3 + x_4 + 0 = 1$$

$$x_1, x_2, x_3, x_4 \geq 0$$

Thus the dual is:

$$\min y_5$$

$$\text{s.t. } 3y_1 + 3y_2 + 3y_3 + 3y_4 + y_5 \geq 0$$

$$y_1 + 4y_2 + 4y_3 + 4y_4 + y_5 \geq 0$$

$$y_1 + 2y_2 + 5y_3 + 5y_4 + y_5 \geq 0$$

$$y_1 + 2y_2 + 3y_3 + 6y_4 + y_5 \geq 0$$

$$-y_1 - 2y_2 - 3y_3 - 3y_4 = -1$$

$$y_1, y_2, y_3, y_4 \geq 0$$

Thus the complementary slackness conditions are as follows:

$$(3x_1 + x_2 + x_3 + x_4 - \gamma)y_1 = 0$$

$$(3x_1 + 4x_2 + 2x_3 + 2x_4 - 2\gamma)y_2 = 0$$

$$(3x_1 + 4x_2 + 5x_3 + 3x_4 - 3\gamma)y_3 = 0$$

$$(3x_1 + 4x_2 + 5x_3 + 6x_4 - 3\gamma)y_4 = 0$$

$$(x_1 + x_2 + x_3 + x_4 - 1)y_5 = 0$$

(Note for the slack $c - A^T y$, since this is equal to 0 I am multiplying by $-1$ to make the copying over a little easier)

$$(3y_1 + 3y_2 + 3y_3 + 3y_4 + y_5)x_1 = 0$$

$$(y_1 + 4y_2 + 4y_3 + 4y_4 + y_5)x_2 = 0$$

$$(y_1 + 2y_2 + 5y_3 + 5y_4 + y_5)x_3 = 0$$

$$(y_1 + 2y_2 + 3y_3 + 6y_4 + y_5)x_4 = 0$$

$$(-y_1 - 2y_2 - 3y_3 - 3y_4 + 1)\gamma = 0$$

**d**

```python
from docplex.mp.model import Model

mdl = Model()

y1 = mdl.continuous_var(name='y1', lb=0)
y2 = mdl.continuous_var(name='y2', lb=0)
y3 = mdl.continuous_var(name='y3', lb=0)
y4 = mdl.continuous_var(name='y4', lb=0)
y5 = mdl.continuous_var(name='y5', lb=-mdl.infinity, ub=mdl.infinity)

mdl.add_constraint(3*y1 + 3*y2 + 3*y3 + 3*y4 + y5 >= 0)
mdl.add_constraint(1*y1 + 4*y2 + 4*y3 + 4*y4 + y5 >= 0)
mdl.add_constraint(1*y1 + 2*y2 + 5*y3 + 5*y4 + y5 >= 0)
mdl.add_constraint(1*y1 + 2*y2 + 3*y3 + 6*y4 + y5 >= 0)
mdl.add_constraint(-y1 - 2*y2 - 3*y3 - 3*y4  == -1)

mdl.minimize(y5)

solution = mdl.solve()

print(mdl.export_as_lp_string())

print(solution)
```

This results in a $y^*$ as follows *rounded to 3 decimals by solver*:

$$y^* = \begin{pmatrix} 0.158 \\ 0.105 \\ 0.070 \\ 0.140 \\ -1.421 \end{pmatrix}$$

We can substitute the values of $y^*$ to our slackness conditions:

$$(3x_1 + x_2 + x_3 + x_4 - \gamma)(0.158) = 0$$

$$(3x_1 + 4x_2 + 2x_3 + 2x_4 - 2\gamma)(0.105) = 0$$

$$(3x_1 + 4x_2 + 5x_3 + 3x_4 - 3\gamma)(0.07) = 0$$

$$(3x_1 + 4x_2 + 5x_3 + 6x_4 - 3\gamma)(0.14) = 0$$

$$(x_1 + x_2 + x_3 + x_4 - 1)(-1.421) = 0$$

$(3(0.158) + 3(0.105) + 3(0.07) + 3(0.14) - 1.421)x_1 = 0$

$\rightarrow\approx (0)x_1 = 0$

$((0.158) + 4(0.105) + 4(0.07) + 4(0.14) + (-1.421))x_2 = 0$

$\rightarrow\approx (0)x_2 = 0$

$((0.158) + 2(0.105) + 5(0.07) + 5(0.14) + (-1.421))x_3 = 0$

$\rightarrow\approx (0)x_3 = 0$

$((0.158) + 2(0.105) + 3(0.07) + 6(0.14) + (-1.421))x_4 = 0$

$\rightarrow\approx (0)x_4 = 0$

$(-(0.158) - 2(0.105) - 3(0.07) - 3(0.14) + 1)\gamma = 0$

$\rightarrow\approx (0)\gamma = 0$

*Note $\approx$ was used to denote any rounding errors*

Any slackness conditions that did not immediately go to 0 can be rewritten as a system of linear equations $Ax^* = b$ where:

$$A = \begin{pmatrix} 3 & 1 & 1 & 1 & -1 \\ 3 & 4 & 2 & 2 & -2 \\ 3 & 4 & 5 & 3 & -3 \\ 3 & 4 & 5 & 6 & -3 \\ 1 & 1 & 1 & 1 & 0 \end{pmatrix}, x^* = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \gamma \end{pmatrix}, b = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Solving this system we get the following steps:

$$\left(\begin{array}{ccccc|c} 3 & 1 & 1 & 1 & -1 & 0 \\ 3 & 4 & 2 & 2 & -2 & 0 \\ 3 & 4 & 5 & 3 & -3 & 0 \\ 3 & 4 & 5 & 6 & -3 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 \end{array}\right) = \left(\begin{array}{ccccc|c} 3 & 1 & 1 & 1 & -1 & 0 \\ 0 & 3 & 1 & 1 & -1 & 0 \\ 0 & 3 & 4 & 2 & -2 & 0 \\ 0 & 3 & 4 & 5 & -2 & 0 \\ 0 & \frac{2}{3} & \frac{2}{3} & \frac{2}{3} & \frac{1}{3} & 1 \end{array}\right) = \left(\begin{array}{ccccc|c} 3 & 1 & 1 & 1 & -1 & 0 \\ 0 & 3 & 1 & 1 & -1 & 0 \\ 0 & 0 & 3 & 1 & -1 & 0 \\ 0 & 0 & 3 & 4 & -1 & 0 \\ 0 & 0 & \frac{4}{9} & \frac{4}{9} & \frac{5}{9} & 1 \end{array}\right)$$

$$\rightarrow= \left(\begin{array}{ccccc|c} 3 & 1 & 1 & 1 & -1 & 0 \\ 0 & 3 & 1 & 1 & -1 & 0 \\ 0 & 0 & 3 & 1 & -1 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & \frac{8}{27} & \frac{19}{27} & 1 \end{array}\right) = \left(\begin{array}{ccccc|c} 3 & 1 & 1 & 1 & -1 & 0 \\ 0 & 3 & 1 & 1 & -1 & 0 \\ 0 & 0 & 3 & 1 & -1 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{19}{27} & 1 \end{array}\right)$$

Thus it is clear from the final line that $\gamma = \frac{27}{19}$ and from the 2nd to last line that $x_4 = 0$

Using this and line 3 that $x_3 = \frac{1}{3}\gamma = \frac{9}{19}$

Then we get $3x_2 = \gamma - x_3 \rightarrow x_2 = \frac{6}{19}$

Lastly $3x_1 = \gamma - x_3 - x_2 \rightarrow \frac{4}{19}$

Therefore we end up with $x^*, y^*$ as follows:

$$y^* = \begin{pmatrix} 0.158 \\ 0.105 \\ 0.070 \\ 0.140 \\ -1.421 \end{pmatrix}, x^* = \begin{pmatrix} \frac{4}{19} \\ \frac{6}{19} \\ \frac{9}{19} \\ 0 \\ \frac{27}{19} \end{pmatrix}$$

We can also see that by strong duality these solutions result in their objective values being $-1.421$ (or $\frac{-27}{19}$ and assuming our objective is $\max -\gamma$) and equal. Thus these solutions are optimal.

## 5

### a

$A = \{(1,2),(2,3),(3,4),(4,2)\}$

$c(A) = \{(1,2),(1,4),(2,3),(2,4),(3,4)\}$

$\min \sum_{h=1}^{4} y_h$

s.t. $x_{1h}, x_{2h}, x_{3h}, x_{4h} \leq y_h, (h = \{1,2,3,4\})$

$x_{1h} + x_{2h} \leq 1, (h = \{1,2,3,4\})$

$x_{1h} + x_{4h} \leq 1, (h = \{1,2,3,4\})$

$x_{2h} + x_{3h} \leq 1, (h = \{1,2,3,4\})$

$x_{2h} + x_{4h} \leq 1, (h = \{1,2,3,4\})$

$x_{3h} + x_{4h} \leq 1, (h = \{1,2,3,4\})$

$\sum_{h=1}^{4} x_{1h} = 1$

$\sum_{h=1}^{4} x_{2h} = 1$

$\sum_{h=1}^{4} x_{3h} = 1$

$\sum_{h=1}^{4} x_{4h} = 1$

$0 \leq x_{Ih}, y_{Ih} \leq 1, (I \in A, h = \{1, 2, 3, 4\})$

# HW 3

## 1

### a

### b

### c

## 2

## 3

## 4

## 5

### a

Omit, Moved to HW4

### b

### c

### d-j

Omit, Moved to HW4